

Building the cyclist-overtake field, and what it revealed

2026-08-28. Card B.1 of docs/czb_work_orders.md. The card asked for a loader, a field check and a perceived-safety readout, on the assumption that the rear-end field would carry over "nearly unchanged". The loader and the checks exist; the assumption did not survive them, and most of this note is about that. Companion documents: docs/lane_entry_note.md (the continuous forms this builds on), docs/czb_validation_roadmap.md §2 (the transfer stage this card serves).

1 What the scenario is, and why it is the right second scenario

The Random design's cyclist overtake gives 2 580 trials from **the same 43 participants** as the cut-in, on 3 clearances (0.5 / 1 / 1.5 m) × 5 truncation points (C1–C5) = 15 cells. It is the only transfer scenario that shares the cut-in's truncated fixed-clip structure: LTAP and truck overtake carry no timepoint at all (they vary criticality, and for LTAP a second speed axis, but have no time series). Comparing a *surface* to the cut-in's surface is therefore possible here and nowhere else, which is why it is first.

2 Two conventions that had to be established from the data

Both are recorded because getting either wrong is silent rather than loud.

The lateral coordinate is Location_Y, not Offset. Both columns are populated and both look reasonable. Location_Y reproduces the study's own criticality labels exactly — edge-to-edge clearance at the passing moment comes out 0.506 / 1.004 / 1.501 m for the traces labelled 0.5 / 1 / 1.5 m. Offset is a *within-lane* coordinate whose reference changes when the ego crosses the lane boundary, and it gives 0.94 / 0.45 / 0.05 m — **inverting** the criticality ordering. A field built on it would run the criticality axis backwards and produce a confident, entirely spurious transfer failure. The label check is asserted in replication/czb/overtake_field_check.py for exactly this reason; it costs nothing and it is the kind of check that pays for itself once.

Roles come from the instructed vehicle, not from lateral span. load_cutin_trace identifies the target as the vehicle that moves sideways. In an overtake that rule is not merely uninformative but **inverted**: here the *ego* pulls out (lateral span 2.24 m) while the cyclist holds its line (0.16 m), so the cut-in rule labels the car as target and the cyclist as ego. That is the mis-assignment recorded as review finding 1.5 in handover_2026-08-27.md, and the mechanism is now identified rather than just observed. comfortzone.overtake.load_overtake_trace assigns by vehicle dimensions instead — the participant drives the car (1.88 m wide), the cyclist is unambiguous at 0.58 m.

3 The onset anchor

The truncation variants Ck need an anchor. The cut-in's convention is $C_k = \text{onset} + 0.3(k-1) \text{ s}$ with C1 ending exactly at manoeuvre onset, so C1 is a genuine pre-event baseline. For the overtake the candidate anchors are the ego's lane-change start and the passing moment, and the data separates them cleanly:

- Under the **onset anchor**, the three clearance conditions are physically *identical* at C1 (clearance spread 0.000 m; they only diverge as the ego's lateral trajectory develops). Observed C1 intervention rates are 0.221 / 0.203 / 0.169 — nearly flat, with a small residual ordering.
- Under a **pass anchor**, C1 would sit 1.2 s before the pass, where the conditions already differ by 0.55 m, predicting a strong C1 ordering that the data does not show.

The onset anchor is adopted. The small residual C1 ordering that remains is the same anticipation signature recorded for the cut-in as query R.1.Q3 — and its appearance here, in a second scenario, upgrades that from a cut-in quirk to a property of the repeated-exposure paradigm.

4 The finding: the field does not represent this scenario's criticality

replication/czb/out/overtake_field_check.md. Over the 15 cells, the field's deficit orders the human intervention rate at Spearman **+0.402**, while a single geometric scalar the preference function does not contain — the clearance the manoeuvre will end with, i.e. the criticality label itself — orders it at **-0.833**.

The reason is structural, and visible in p_lane rather than inferred. The lane-entry weight answers "does the car-following conflict geometry apply?", and in this scenario the answer is *yes, fully*, for the whole response window in all three conditions: P_lane never leaves 0.843–1.000. The scenario's manipulated variable is invisible to it.

Two candidate repairs were built and tested, both flag-gated and both defaulting off:

- `lane_entry_bidirectional` — the released form and the 2026-08-27 continuous form both clamp the lateral projection to *inward* motion, so an object moving laterally away is treated as stationary. That is harmless in a cut-in and discards the entire signal in an overtake. Projecting symmetrically over-corrects instead: `P_lane` collapses to 0 by C3 in every condition, because linear extrapolation of a still-developing lateral move over a ~2.2 s closure says the ego will have cleared comfortably. It ends up ordering the cells *negatively* (−0.537). Kept, default off, as a tested-and-rejected variant.
- `lane_entry_shape_k` — the S-shaped ramp (§5). It changes nothing here, because a shape function applied to a saturated input is a no-op.

The honest conclusion is that neither the overlap fraction nor its projection is the right functional for this scenario. What varies across the conditions is not *whether* overlap will occur — it will not, in any of them — but the size of the **lateral comfort margin** at the pass. Passing a cyclist at 0.5 m is uncomfortable precisely while being uncontroversially collision-free, which is a comfort quantity the preference function's collision-oriented lateral machinery does not carry. Adding a lateral-clearance comfort term is a change to the preference function's structure and therefore a review decision, not one this card takes.

5 The S-shaped lane-entry ramp

Jonas's proposal of 2026-08-28: the ramp from no overlap to full overlap is currently linear, and it should plausibly be a sigmoid — a sliver of predicted overlap does not yet feel like a rear-end conflict, but once overlap is established the situation becomes one quickly. Implemented as `lane_entry_shape(u; k)`, a normalized logistic with $g(0) = 0$, $g(1) = 1$ exactly for every k and $g(u; 0) = u$, so the published linear ramp is nested at $k = 0$ and every existing number is untouched by default.

Swept on the cut-in surface, where the overlap fraction actually spans its range
(`replication/czb/out/lane_entry_shape_check.md`):

k	0 (linear)	+8	+12	+20	+100 (≈ step)	−8
RMSE	0.1189	0.1156	0.1154	0.1230	0.1457	0.1211

The direction of the proposal is supported: positive k fits better, negative k (the reflected shape) fits worse, and there is a genuine interior optimum at $k \approx 12$. The improvement is modest — 2.9% of the linear form's error — so the shape is real but not dramatic on this design. A useful by-product: the $k \rightarrow \infty$ limit is a step at half overlap, which is effectively the released binary gate, and it is clearly the worst of the family. That independently corroborates the 2026-08-27 decision to make lane entry continuous — the gain there was not an artifact of choosing a linear ramp in particular.

Whether to fit k is not settled here. Fitting it would make k the first fitted parameter upstream of the boundary, and the stage-0 result draws much of its force from the field having no fitted constants at all. One shape parameter does not destroy that, but it must be reported as a field parameter, and for any transfer test it would have to be frozen at its cut-in value before a transfer scenario is scored. Raised as a query.

6 The transfer result, and the baseline correction it depended on

`replication/czb/out/transfer_overtake_summary.md`. Fitting stage 1 on the cut-in (population median 5 352, between-driver σ 0.209, response sd 970) and scoring the 15 overtake cells:

model	RMSE
frozen — nothing refitted (the roadmap's primary)	0.176
free lapse only — boundary level and spread frozen	0.129
free lapse + one level shift (the roadmap's secondary)	0.110
full refit (a ceiling, not a transfer model)	0.113
<i>chance</i> — <i>the overtake's grand mean</i>	<i>0.158</i>

The pre-onset intervention rate is 0.081 in the cut-in and 0.198 in the overtake — 2.4× higher before either manoeuvre develops. Frozen transfer is therefore worse than chance for a reason that has nothing to do with the comfort boundary. Freeing **only the lapse** — which the C1 cells identify on their own, and which by construction carry no boundary information, since C1 ends where the field is zero — brings the model comfortably past chance and to

within 0.016 of the full-refit ceiling. The boundary level and spread fitted on the cut-in do most of the work in the overtake, once the paradigm baseline is matched.

The recommendation for card B.4 follows: the primary transfer analysis should freeze the boundary and free each scenario's lapse, rather than freeing nothing. This is a weaker concession than the roadmap's secondary "one uniform shift per scenario", which absorbs the boundary itself and makes the test close to vacuous — visible above, where the level shift buys 0.019 more and slightly beats the full refit.

7 The field-independent check that bounds all of this

replication/czb/out/cross_scenario_consistency.md. Because all 43 participants saw all four Random scenarios, the one-scalar claim's central prediction can be tested with no field at all: a driver who sits low in one scenario should sit low in the others.

Per-driver criticality-adjusted propensity, with split-half reliabilities of 0.93–0.98 (so the measure is clean), correlates across scenario pairs at **+0.50 to +0.74**, which is **0.53 to 0.78 of the reliability ceiling**, mean **0.69**.

About 69% of the reliable per-driver signal is shared across scenarios. That is substantial positive evidence for a common trait — the strongest non-field evidence available for the one-scalar framing — and it simultaneously sets an upper bound: roughly a third of reliable per-driver variation is scenario-specific, so no one-scalar transfer model can explain everything, and B.4 should be scored against that bound rather than against perfection.

8 What this card did not do

The ordered comfort/dread model is **not** fitted on the overtake. The study's third question is scenario-dependent, and its Random-design wording for the cyclist overtake is recorded as undocumented and "to be confirmed" in the dataset's own context file; the column carries 0/1 here against the cut-in's 0/1/2. The intervention response (CZB_1) is identically defined in both scenarios and is what transfers; the comfort/dread pair remains cut-in-only until the question wording is confirmed.